higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(First Paper)
NQF LEVEL 2

21 FEBRUARY 2019

SYMBOL	EXPLANATION
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD/RM	Reading from a table/graph/drawing/document/map
F	Choosing correct formula
SF	Substitution in formula
MF	Manipulation of formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off
E	Explanation

This marking guideline consists of 8 pages.

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QUESTION 1 *Do not deduct marks if the 'R' sign is omitted.				
QUESTION		**SOLUTION**	**EXPLANATION**	
1.1	1.1.1	(6 x 20) – 8 ÷ 2 = 26✓ – 4✓ = 22✓ **(answer only - one mark)**	 2MA 1A	 (3)
	1.1.2	½ of 200 – (–10) = 100✓ + 10✓ = 110✓ **(answer only - one mark)**	 2M 1A	 (3)
1.2		89 ; 36 ; 0 ; -66 ✓ ✓	2A 1 mark per 2 correct order	(2)
1.3		$\frac{102}{186} \times 100$ ✓ = 54,84%✓ ≈ 55✓%	1M 1A 1C (3 Answer only)	(3)
1.4		6 300 ÷ 1 000✓ 6,3 ✓kg	1M 1A (2 Answer only)	(2)
1.5		$\frac{15}{100} \times R\,10\,200$ = R1530✓ $\therefore R\,10\,200 - R\,1\,530$ = R8 670✓ **OR** $\frac{85}{100} \times R\,10\,200$ = R8 670✓	1M 1A 1CA (3 Answer only) 2MA 1CA (3 Answer only)	(3)
1.6		Wandile: $\frac{1}{5} \times R\,6\,000$✓ = R1 200✓ Gladys: $\frac{4}{5} \times R\,6\,000$ = R4 800✓	1M 1A (2 Answer only) 1M 1A (2 Answer only)	(4)

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1.7	1 h 35 + 1 h 16✓ + 2 h 23 4 h 74 ∴ 5 hours✓ 14 minutes✓	1M 2A (3 Answer only)	(3)
1.8	**Makro:** $\frac{R\,162{,}95}{24} = R\,6{,}79 \;\; per \;\; can$ **Pick & Pay:** $\frac{R\,34{,}99}{6} = R\,5{,}83 \;\; per \;\; can$ Pick & Pay is more economical/cheaper.✓ Or any other suitable way of calculation.	1M 1A (3 Answer only) 1A 1R (reason only – 1 mark maximum)	(4)
1.9	2:24 pm✓ **(Must have pm, otherwise mark as incorrect)**	1A	(1)
1.10	0,32 × 100✓ = 32%✓	2MA(2 Answer only)	(2)
			[30]

QUESTION 2				
QUESTION		**SOLUTION**	**EXPLANATION**	
2.1	2.1.1	B✓✓	2RD	(2)
	2.1.2	D✓✓	2RD	(2)
	2.1.3	A✓✓	2RD	(2)
	2.1.4	C✓✓	2RD	(2)
2.2	2.2.1	Size of screen = $\sqrt{28^2 + 15{,}7^2}$ = $\sqrt{784 + 246{,}49}$ = $\sqrt{1030{,}49}$✓ = 32,10✓ ≈32✓inches	2SF 2A 1R (5 Answer only)	(5)

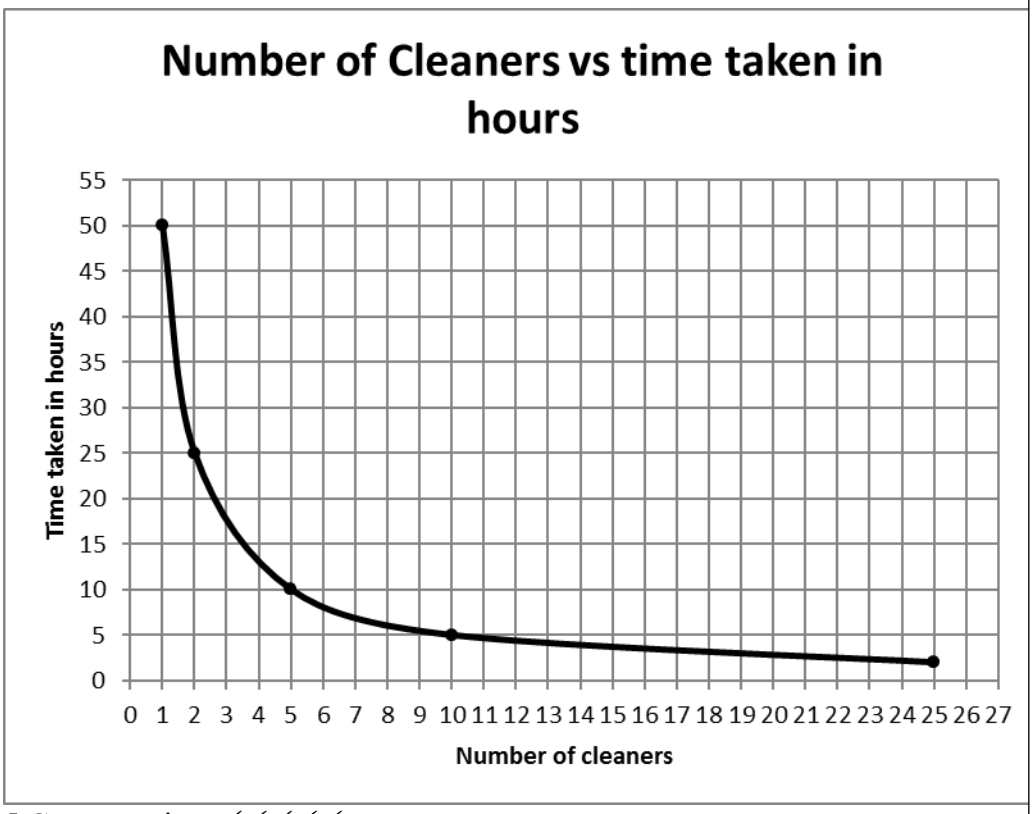
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	2.2.2	Area = length × breadth = 28✓× 15,7✓ = 439,6 inches²✓ (or 439,60)	2SF 1A (Answer with unit) (3 Answer only – with unit)	(3)
2.3	2.3.1	Volume = length × breadth × height = 412✓ × 201✓× 122✓ = 10 103 064 cm³✓	3SF 1A(Answer with unit) (4 Answer only- with unit)	(4)
	2.3.2	No✓ The length of the pool is 412 cm or 4,12 m. The longest side of the yard is 4 m which indicates that it is impossible for the pool to fit into the yard regardless of the orientation of the pool.✓✓ Or accept any other similar answer.	1A 2J	(3)
2.4	2.4.1	1 : 50 103 : x x = 50✓ × 103✓ = 5150 ✓cm $\therefore \frac{5\ 150}{1\ 000} = 5{,}15\ \ km$	 2M 1A 1C 1CA (5 Answer only)	(5)
	2.4.2	No.✓ The map indicates 4 x 4 only.✓	1A 1J	(2)
				[30]

QUESTION 3		***Do not deduct marks if the 'R' sign is omitted.**		
QUESTION		**SOLUTION**	**EXPLANATION**	
3.1	3.1.1	Ben Stokes✓	1RT	(1)
	3.1.2	01/11/2018 or 1st of November 2018✓	1RT	(1)
	3.1.3	DH130✓	1RT	(1)
	3.1.4	R115✓ × 6✓ = R690	2MA	(2)
	3.1.5	$\frac{100}{115} \times R1\,899$ $= R1\,651{,}30$	2M 1A (3 Answer only)	(3)

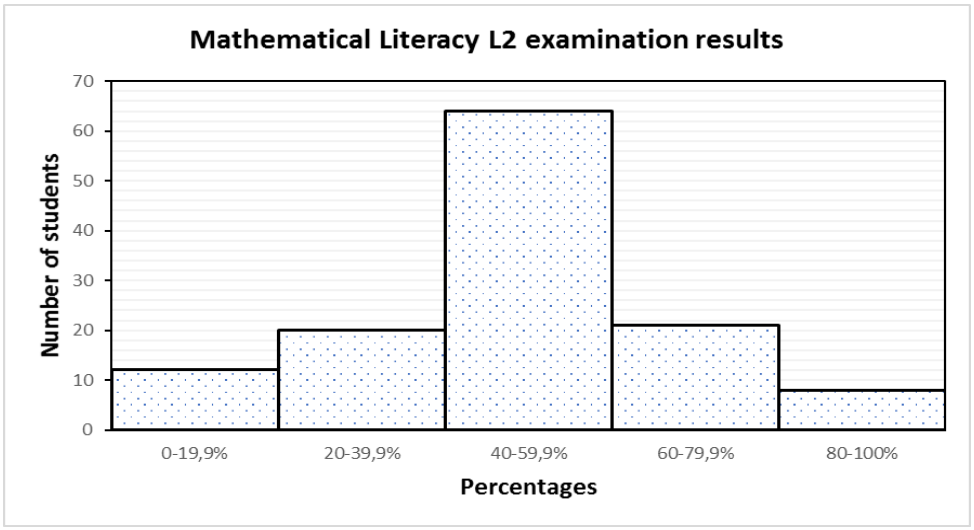
	3.1.6	No✓ The quote is valid for 14 days which means that after the 15th the quote is no longer valid. ✓ Or any other suitable reason.	1A 1R/J	(2)
3.2	3.2.1	Commission from sales✓	1RT	(1)
	3.2.2	Rent or Car repayment or Insurance✓ (Any one of the above)	1RT	(1)
	3.2.3	A: R5 000 + R7 100✓ =R12 100✓ B: R12 100 – R11 015✓ = R1 085✓	1M 1A (2 Answer only) 1M (CA answer of A) 1A (2 Answer only)	(4)
	3.2.4	Surplus✓	1A	(1)
	3.2.5	Groceries✓ Or Entertainment✓	1RT	(1)
3.3	545 km – 400 km = 145 km travelled extra✓ Cost = (2 × R465)✓ + (R3,85 × 145)✓ = R930 + R558,25 = R1 448,25✓		1A 2MA 1A	(4)
3.4	3.4.1	R4,20✓	1RT	(1)
	3.4.2	-14,3✓% (No mark for 14,3%)	1RT	(1)
	3.4.3	It means that there was a decrease in the bank costs.✓	1R/J	(1)
	3.4.4	Cost = R4,00✓ + $(\frac{1,3}{100} \times R1\,500)$ = R4,00 + R19,50 = R23,50✓ (If R23,5 give a maximum of 3 marks)	1F 2M 1A (4 Answer only)	(4)
	3.4.5	Decreased✓	1A	(1)
				[30]

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QUESTION 4		***Do not deduct marks if the 'R' sign is omitted.**		
QUESTION		**SOLUTION**	**EXPLANATION**	
4.1	4.1.1	32✓ ; 40✓	2A	(2)
	4.1.2	96✓ ; 192✓	2A	(2)
	4.1.3	Constant difference✓	1A	(1)
4.2	4.2.1	A: 50 ÷ 5✓ = 10 hours✓ B: 50 ÷ 5✓ = 10 cleaners✓	2MA (2 Answer only) 2MA (2 Answer only)	(4)
	4.2.2	Number of cleaners✓	1A	(1)
	4.2.3	Time taken to clean✓	1A	(1)
	4.2.4	Indirect or Inverse✓ As the number of cleaners increase,✓ the hours to complete the job will decrease.✓	1A 2R/J	(3)
	4.2.5	 5 Correct points ✓✓✓✓✓ 1 Curve✓ 1 Heading ✓ 1 Axes (both correct) ✓ - No marks if one axis is incorrect		(8)
	4.2.6	Accept answers from 3 to 4 hours✓ (do not except 3,33 hours)	1RT	(1)

4.3	4.3.1	R100✓	1RT	(1)
	4.3.2	20 ✓handbags	1RT	(1)
	4.3.3	A: 1000 ÷ 100✓ = 10 ✓handbags B: 15 × 100✓ = R1 500✓	2MA (2 Answer only) 2MA (2 Answer only)	(4)
	4.3.4	Direct relationship✓	1A	(1)
				[30]

QUESTION 5				
QUESTION		**SOLUTION**	**EXPLANATION**	
5.1	5.1.1	Questionnaire✓ The SSS used the registration forms to gather information. ✓	1A 1R/J	(2)
	5.1.2	A = 12✓ B = 8✓	2RT	(2)
	5.1.3	C = 𝍸 𝍸 𝍸 𝍸✓ D = 𝍸✓	1A 1A	(2)
	5.1.4	E = 75✓✓	1RT 1A (CA 75 from 5.1.2)	(2)
	5.1.5	2✓	1RT	(1)
	5.1.6	$\frac{3}{75} \times 100$ = 4✓%	2M (CA 75 from 5.1.4) 1A (3 Answer only)	(3)
5.2	5.2.1	85,44; 86,38; 87,52; ✓ 88,26; 90,02; 91,20✓	2RT (1 mark for every 3 correct)	(2)
	5.2.2	260 ÷ 100✓ = 2,6 ✓m	1M 1C (2 Answer only)	(2)
	5.2.3	$\frac{87,52 + 88,26}{2}$ = 87,89✓ m	2M 1A (3 Answer only)	(3)
	5.2.4	91,20 ✓m OR The second throw of the series.✓	1RT	(1)
	5.2.5	104,8– 91,20✓ = 13,6 m✓	1MA 1CA with U (2 Answer only)	(2)

5.3	 5 Correct bars ✓✓✓✓✓ 1 Heading ✓ Histogram ✓ 1 horizontal and vertical axes (both correct) ✓	(8)
		[30]
	TOTAL:	**150**

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